
Housing Price Prediction: A Comparative Study of Regularization Techniques

Miguel Nino Adalla (mva54) Reuben Thomas (rmt135) Nicole Krivokapic (nk752)

1 Executive Summary

This project investigates linear regression models, enhanced through regularization, to predict housing prices using the Ames, Iowa Housing dataset. We systematically compared standard Linear Regression, Ridge Regression (L2), and Lasso Regression (L1), each trained on two datasets: one with and one without extensive feature engineering. The goal was to evaluate the impact of regularization and feature engineering on prediction accuracy.

The key findings include:

- Ridge Regression with feature engineering was the best model, achieving $R^2 = 0.937$ and RMSE = 0.108.
- Feature engineering reduced test RMSE by up to 10% and increased R^2 by up to 1.8%.
- The most predictive features included “TotalSF_Log”, “OverallQual”, “GrLivArea_Log”, and “Neighborhood_Price”.
- Regularization effectively addressed multicollinearity and improved generalization.

The developed models can assist various stakeholders in the real estate market, including homebuyers, sellers, and investors, by providing data-driven estimates of property values. Future work could explore more complex modeling approaches and incorporate additional external data sources to further improve prediction accuracy.

2 Introduction

This project addresses the problem of predicting housing prices in Ames, Iowa, through supervised learning. Specifically, we aim to determine how well linear models with regularization can estimate sale prices, and how feature engineering would influence their performance.

3 Motivation

This project is important because accurate housing price prediction helps various stakeholders make informed decisions. Potential homebuyers can determine if a house is fairly priced, sellers can set competitive prices, and investors can identify properties with growth potential. Understanding the factors that influence housing prices can inform urban planning and development strategies.

I’m particularly excited about this project because it provides an opportunity to apply statistical learning techniques to a real-world problem with tangible implications. Real estate is one of the largest markets in the economy, and data-driven approaches can help make this market more efficient and transparent.

Existing questions in this area include:

- Which features have the strongest correlation with housing prices?

- How do different neighborhoods impact property values?
- Can regularization techniques improve prediction accuracy and model interpretability?
- Does feature engineering improve accuracy and interpretability?

4 Method

4.1 Dataset

For this project, I used the Ames Iowa Housing dataset, which contains information on residential properties sold in Ames, Iowa between 2006 and 2010. This comprehensive dataset includes 79 explanatory variables describing various aspects of residential homes. The data is in tabular format with both numerical and categorical features. Key feature categories include:

1. Property characteristics: Square footage, number of rooms, bathrooms, etc.
2. Location information: Neighborhood, zoning classification
3. Quality and condition ratings: Overall quality, exterior condition, etc.
4. Time-related features: Year built, year remodeled, year sold
5. Special features: Fireplaces, pools, garages, etc.

4.2 Problem Definition & Model Selection

Because during EDA we observed that SalePrice was slightly skewed right, we chose to add a new log-transformed feature, SalePrice_Log, to use as the target variable. Our goal was to train three regression models to predict SalePrice_Log. We implemented these three models, each on two datasets:

1. Standard Linear Regression: Provides a baseline performance using Ordinary Least Squares (OLS)
2. Ridge Regression: Applies L2 regularization to reduce model complexity and mitigate multicollinearity
3. Lasso Regression: Employs L1 regularization which performs feature selection by shrinking less important feature coefficients to zero

Each model was evaluated on both raw and feature-engineered datasets, for a total of six experiments. This approach allowed us to isolate the effect of feature engineering and regularization.

4.3 Implementation Steps

1. Exploratory Data Analysis (EDA):
 - Analyze feature distributions
 - Identify correlations between features and target variable
 - Detect any outliers and anomalies, if any
2. Data Preprocessing:
 - Handle missing values through imputation or removal
 - Transform skewed numerical features using log or other transformations
 - Encode categorical variables using one-hot encoding or label encoding
3. Feature Engineering:
 - Create new features that might have predictive power
 - Apply feature scaling to normalize the range of features
4. Model Training:
 - Split data into training and testing sets
 - Train three different linear regression models each with both dataset types
 - Tune hyperparameters using cross-validation

5. Model Evaluation:
 - Assess performance on test set
 - Compare models using various metrics
 - Analyze the improvement in using feature engineering for each of the three models
6. Predictions Visualizations:
 - Analyze predictions of the best performing model
 - Visualize relationships between actual and predicted sales prices
 - Visualize residual distribution between actual and predicted sales prices
7. Feature Importance:
 - Compare feature importance across different models
 - Visualize the most influential features using bar charts
8. Conclusion

4.4 Evaluation Metrics

To evaluate model performance, we used the root mean squared error (RMSE), mean absolute error (MAE), R squared score (R²), cross-validation scores (5 fold).

5 Insights and Methodology Rationale

5.1 Exploratory Data Analysis Insights

5.1.1 Target Variable Transformation

We chose to log-transform the SalePrice target variable because the original distribution exhibited significant right skewness (1.74 skewness value). This transformation helped reduce this positive skewness, and create a more normal distribution. This is important for linear models which assume that the target variable is normally distributed.

5.1.2 Key Correlation Findings

The correlation analysis revealed several features with strong positive correlations to SalePrice, namely OverallQual, GrLivArea, GarageCars, and TotalBsmtSF. This helped us determine which features will likely be most predictive and influential for the models. It also helped guide our feature engineering efforts, particularly in creating composite features that amplified these relationships.

5.1.3 Categorical Feature Analysis

The boxplot analysis of categorical variables like Neighborhood and ExterQual revealed significant group-level pricing patterns. Some neighborhoods consistently showed higher median prices, while quality metrics for kitchens and exteriors demonstrated substantial price impact across quality levels. This confirmed the need for effective encoding strategies for these categorical variables.

5.2 Preprocessing Rationale

5.2.1 Missing Value Strategy

We needed to rigorously handle missing values because our models would not be able to handle them if kept in the dataset. At the same time, we didn't want to sacrifice any critical information. Therefore, our missing value imputation approach was carefully tailored to each feature type.

- For categorical features like FireplaceQu, missing values were replaced with 'None' to preserve the semantic meaning (absence of a fireplace)
- For numerical features like LotFrontage, we employed group-based imputation using neighborhood medians, providing more accurate estimates than global measures

This approach prevented information loss while maintaining data integrity for model training

5.2.2 Feature Transformation & Encoding

We applied log transformations to heavily right-skewed features like LotArea, MiscVal, and BsmtFinSF1. By doing so we make their relationships more linear, reduce the influence of outliers, and ultimately improve the performance of our linear models. Our encoding approach was feature-specific:

- Ordinal encoding was applied to quality features (ExterQual, KitchenQual, etc.) to capture their inherent ordering (Excellent > Good > Average, etc.)
- One-hot encoding was used for nominal categorical features with low cardinality, allowing models to learn group differences without imposing artificial ordering
- These encoding choices allowed our models to effectively utilize both ordered and unordered categorical information

5.3 Feature Engineering Rationale

5.3.1 Domain-Informed Transformations

Our feature engineering decisions were driven by domain knowledge of real estate valuation and by our prior observations obtained during EDA. Some important new features include:

- Time-Based Features (eg. HouseAge, IsRemodeled): Captures potential value added through renovation or lost through depreciation. These are often not linearly related to age.
- Area and Size Features (eg. TotalSF, TotalBath): Created to provide a more comprehensive size metric than any individual area measurement.
- Ratio and Proportion Features (e.g. LotRatio, BedroomRatio): Captures relationship between more relative relationships that might be missed by linear models.
- Products and Aggregation of Features (eg. OverallQualCond, Neighborhood_Price): Captures more general aspects of homes that could have a combined effect beyond their individual impact.

5.4 Model Selection and Hyperparameter Tuning

5.4.1 Regularization Approach

We specifically chose Ridge and Lasso regression models to address challenges in the housing dataset:

- The high-dimensional feature space (310 features after engineering) created overfitting risks
- Multicollinearity between features (e.g., various area measurements) could destabilize standard linear regression
- Ridge (L2) regularization shrinks all coefficients to reduce model complexity
- Lasso (L1) regularization performs feature selection by reducing some coefficients to zero

5.4.2 Hyperparameter Selection

Our GridSearch explored a range of alpha values:

- Ridge: [0.001, ..., 1000]
- Lasso: [0.0001, ..., 10]

This wide range was deliberately chosen to explore the spectrum from under-regularization to over-regularization, allowing us to identify the optimal balance between model fit and generalization.

6 Results

6.1 Model Performance

After training and evaluating all the models, I found that the regularized models performed better than the standard linear regression. Ridge regression with feature engineering showed the best overall

performance with an RMSE of 0.108 and R^2 of 0.937, closely followed by Lasso regression (FE) with similar metrics.

6.2 Performance Metrics Analysis

Model	RMSE (Test)	MAE (Test)	R^2 (Test)	CV RMSE
Linear Regression (No FE)	0.1330	0.0905	0.9043	0.1582
Ridge Regression (No FE)	0.1182	0.0798	0.9245	0.1407
Lasso Regression (No FE)	0.1202	0.0807	0.922	0.1435
Linear Regression (FE)	0.1161	0.0830	0.9270	0.1418
Ridge Regression (FE)	0.1076	0.0749	0.9374	0.128
Lasso Regression (FE)	0.108	0.0755	0.937	0.1269

Table 1: Performance metrics across all models

Feature Engineering Benefits:

- Ridge: $\sim 9\%$ RMSE reduction, $+1.4\%$ R^2
- Lasso: $\sim 10\%$ RMSE reduction, $+1.6\%$ R^2
- Linear: $\sim 13\%$ RMSE reduction, $+2.5\%$ R^2

6.3 Prediction Visualization Analysis

6.3.1 Actual vs. Predicted Analysis

The comparison of actual vs. predicted values showed that the best model (Ridge Regression with feature engineering) demonstrated very tight clustering along the identity line, indicating very strong predictive accuracy. The model performed consistently across most of the price range. However there looked to be a slight tendency to under predict for the highest-value homes, suggesting potential for improvement in this region.

6.3.2 Residual Analysis

The residual plots provided additional diagnostics. The scatterplot confirmed that the model was very consistent, and that it had a tendency to underpredict high-value properties. The distribution was approximately normal with a very slight right skew, again confirming the underprediction tendency.

6.4 Feature Importance Insights

The feature importance analysis identified the most influential predictors:

- TotalSF_Log: Emerged as the single most important feature, confirming that total home size is the primary value driver
- OverallQual: The subjective quality rating showed very high importance, highlighting the value of expert assessment
- GrLivArea_Log: Above-ground living area remained highly influential even with total square footage present
- Neighborhood_Price: This engineered feature effectively captured location value, demonstrating the success of our feature engineering approach

Between the the Ridge and Lasso models, both approaches identified similar top features. TotalSF_Log was the most important feature in both models, showing the strongest positive coefficient. However, there were some interesting differences. Lasso tended to prefer a few, very important features, whereas Ridge regression distributed importance across more features. This is characteristic of L1 and L2 regularization, and confirms that our models behaved normally.

7 Discussion

7.1 Expected vs. Actual Results

We expected that both Ridge and Lasso regression would outperform OLS due to their ability to handle multicollinearity and prevent overfitting. We anticipated that Lasso might perform particularly well given its feature selection capability in a dataset with numerous features.

Our actual results largely aligned with these expectations. Both regularized models outperformed the standard linear regression, confirming the value of regularization for this problem. However, Ridge regression slightly outperformed Lasso regression. This suggests that feature shrinkage (reducing the magnitude of all coefficients) was more beneficial than feature selection (eliminating some features entirely) for this particular dataset.

The feature importance analysis revealed some surprising insights. While we expected features like overall quality and above-ground living area to be significant predictors, we did not anticipate the substantial impact of basement-related features. The finished basement area emerged as one of the top predictors, indicating that below-ground living space is highly valued in this market.

7.2 Limitations and Future Work

Despite the strong performance of our models, several limitations should be acknowledged:

1. Temporal relevance: The dataset only includes houses sold between 2006-2010, potentially limiting generalizability to current market conditions, especially considering the significant fluctuations in the housing market since that period.
2. Geographic specificity: The models are trained on data from Ames, Iowa only and may not transfer well to other real estate markets with different valuation dynamics.
3. Missing external factors: Economic indicators, interest rates, and market trends are not captured in the dataset but likely influence housing prices significantly.
4. Model simplicity: While linear models offer interpretability, they may miss complex non-linear relationships between features and price.

To improve this analysis in future work, I would recommend:

1. Advanced modeling techniques: Implement ensemble methods or gradient boosting models to capture more complex patterns and potentially improve prediction accuracy.
2. Time series analysis: Incorporate temporal trends in housing prices to account for market fluctuations and seasonality effects.
3. External data integration: Include economic indicators, school district ratings, crime statistics, and neighborhood development data to provide additional context.
4. Neural network approach: Explore deep learning techniques for potentially higher accuracy, especially for capturing non-linear relationships in the data.

8 Conclusion

In conclusion, this project demonstrated the effectiveness of regularized linear regression techniques and feature engineering for housing price prediction. Both Ridge and Lasso provided improvements over standard linear regression, with Ridge regression showing the best overall performance. The analysis highlighted the importance of overall quality, living area size, basement features, and neighborhood in determining housing prices, providing valuable insights for real estate stakeholders, homebuyers, and sellers in the Ames market.

The regularization approach successfully addressed multicollinearity issues common in housing data and prevented overfitting. The interpretability of the linear models allowed for clear identification of key value drivers, making the findings actionable for various stakeholders in the real estate market.

Future research directions could explore more complex modeling techniques and incorporate additional data sources to further enhance prediction accuracy, particularly for luxury properties where the current models showed slightly lower performance.